QMIS ERP — Full Module Flow Test Guide

Purpose: Test **every sidebar module** in campus order: what you do → where data is stored → who sees it next.

Updated: 18 October 2026

Stack: Postgres 18 @5433 · API http://127.0.0.1:8000 · App http://localhost:5173

Status sheet: reports/QMIS-ERP-Role-Module-Status.480-complete.17oct.docx (480/0/0/0)

Regenerate PDF: python3 reports/_build_module_flow_test_guide_pdf.py

How to use this guide

- Start API (cd backend && ./run-local.sh) and frontend (npm run dev).
- Follow roles in **campus order** (sections below). Later roles need data created earlier (classes, students, employees, logins).
- For each module row:
- **Do** — actions on the screen
- **Stored in** — API / collection
- **Reflects to** — other roles or screens that must show the same data
- Pass rule: after create/edit, **hard refresh (F5)** — data still present (unless noted as demo-only / external).
- If a staff role login is missing: Administrator / HR → create employee + login, then continue.

Cross-campus data flow (summary)

```
flowchart LR
Admin[Administrator masters + users]
HR[HR employees + leave]
PRM[PRM admissions + FO passes]
Fin[Account Head fees]
Prin[Principal academics]
Teach[Teacher attendance marks LP]
Parent[Parent Student portals]
Ops[Ops Library Gate Stores Transport]
Audit[JD Audit + Process Quality]
Admin --> HR
Admin --> PRM
Admin --> Fin
Admin --> Prin
HR --> Teach
PRM --> Parent
Prin --> Teach
Teach --> Parent
Admin --> Ops
Ops --> Audit
```

Seed logins (development)

· Role · Email · Password ·

· Administrator · admin@qmis.edu · admin123 ·

· Super Admin · superadmin@qmis.edu · superadmin123 ·

· HR · hr@qmis.edu · hr12345 ·

· Account Head · accounthead@qmis.edu · accounts123 ·

· PRM (Front Office) · prm@qmis.edu · prm12345 ·

· Principal · principal@qmis.edu · principal123 ·

· Director · director@qmis.edu · director123 ·

· Coordinator · coordinator@qmis.edu · coord123 ·

· Teacher · teacher@qmis.edu · teacher123 ·

· Parent · (from Admission enroll) · (set at enroll) ·

· Student · (optional student login) · (if provisioned) ·

· Librarian · (HR create staff login / seed if present) · — ·

· Gate Keeper · (HR create staff login) · — ·

· Gate Keeper Manager · (HR create staff login) · — ·

· Canteen Manager · (HR create staff login) · — ·

· IT Support Manager · (HR create staff login) · — ·

· Stationery Store Manager · (HR create staff login) · — ·

· Housekeeping Manager · (HR create staff login) · — ·

· Transport Manager · (HR create staff login) · — ·

· Driver · (HR create staff login) · — ·

· Joint Director · (HR create staff login) · — ·

· Joint Director Assistant · (HR create staff login) · — ·

· Joint Director Audit · (HR create staff login) · — ·

· Process Auditor · (HR create staff login) · — ·

· Quality Auditor · (HR create staff login) · — ·

Override passwords via backend/.env *_SEED_PASSWORD if set.

Step 1 — Administrator

Login: admin@qmis.edu / admin123

Modules in sidebar: 26

1.1 Dashboard

1.1.1 Dashboard

- **Path:** /dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** GET /api/v1/admin/dashboard — Live KPIs from campus aggregates
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/admin/dashboard
- **Pass:** List/detail still correct after F5; no console/API 500.

1.2 Admissions

1.2.1 Admission Enquiry

- **Path:** /admin/front-office/admission-enquiry
- **Do:** Open **Admission Enquiry** (/admin/front-office/admission-enquiry). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Student Database; Parent login; class lists
- **Pass:** List/detail still correct after F5; no console/API 500.

1.2.2 Admission List

- **Path:** /admin/front-office/admission-list
- **Do:** Open **Admission List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Student Database; Parent login; class lists
- **Pass:** List/detail still correct after F5; no console/API 500.

1.3 User Database

1.3.1 Student Database

- **Path:** /admin/user-management/student-database
- **Do:** Open **Student Database** (/admin/user-management/student-database). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/students|/admissions — Campus student SoT; Parent/Student portals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/students|/admissions
- **Pass:** List/detail still correct after F5; no console/API 500.

1.3.2 Employee Database

- **Path:** /admin/user-management/employee-database
- **Do:** Open **Employee Database** (/admin/user-management/employee-database). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

1.4 Class Details

1.4.1 Class Details

- **Path:** /admin/class/class-details
- **Do:** Open **Class Details**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/masters (classes/sections) — Teachers mark attendance against these classes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/masters (classes/sections)
- **Pass:** List/detail still correct after F5; no console/API 500.

1.5 Attendance

1.5.1 Students

- **Path:** /admin/attendance/students-list
- **Do:** Open **Students**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/attendance — Class attendance; Parent/Student can read published
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

1.5.2 Employees

- **Path:** /admin/attendance/employees-list
- **Do:** Open **Employees**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/attendance — Class attendance; Parent/Student can read published
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

1.6 Activities

1.6.1 Cultural

- **Path:** /admin/activities/cultural-list
- **Do:** Open **Cultural** (/admin/activities/cultural-list). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

1.6.2 Sports

- **Path:** /admin/activities/sports-list
- **Do:** Open **Sports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

1.6.3 Chess Result

- **Path:** /admin/activities/chess-result
- **Do:** Open **Chess Result** (/admin/activities/chess-result). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Parent/Student Exam Result after publish
- **Pass:** List/detail still correct after F5; no console/API 500.

1.6.4 Competitions

- **Path:** /admin/activities/competitions-list
- **Do:** Open **Competitions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

1.7 Documents

1.7.1 Student Documents

- **Path:** /admin/documents/student-documents
- **Do:** Open **Student Documents**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/students|/admissions — Campus student SoT; Parent/Student portals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/students|/admissions
- **Pass:** List/detail still correct after F5; no console/API 500.

1.7.2 Employee Documents

- **Path:** /admin/documents/employee-documents
- **Do:** Open **Employee Documents**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

1.8 Task Management

1.8.1 Assign Tasks

- **Path:** /admin/task-management/assign-tasks
- **Do:** Open **Assign Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

1.8.2 Assigned Tasks

- **Path:** /admin/task-management/assigned-tasks
- **Do:** Open **Assigned Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

1.9 Leave Request

1.9.1 My Leave Requests

- **Path:** /admin/leave-request/my-requests
- **Do:** Open **My Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.

- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

1.9.2 Received Leave Requests

- **Path:** /admin/leave-request/received
- **Do:** Open **Received Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

1.10 Announcement

1.10.1 Announcement

- **Path:** /admin/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

1.11 TC Request Approval

1.11.1 TC Request Approval

- **Path:** /admin/tc-request-approval
- **Do:** Open **TC Request Approval** (/admin/tc-request-approval). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/tc-requests — Parent/Student file → PRM forward → Admin decide
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tc-requests
- **Pass:** List/detail still correct after F5; no console/API 500.

1.12 Communication

1.12.1 Communication

- **Path:** /admin/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

1.13 Calendar

1.13.1 Calendar

- **Path:** /admin/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

1.14 Notifications

1.14.1 Notifications

- **Path:** /admin/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

1.15 Escalation Management

1.15.1 Escalation Management

- **Path:** /admin/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

1.16 RBAC

1.16.1 User Creation

- **Path:** /admin/rbac/user-creation
- **Do:** Open **User Creation** (/admin/rbac/user-creation). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students
- **Pass:** List/detail still correct after F5; no console/API 500.

1.16.2 Roles

- **Path:** /admin/rbac/roles
- **Do:** Open **Roles**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** RBAC / permissions matrix — Controls which APIs each role can call
- **Reflects to:** Same role after refresh; related roles that share RBAC / permissions matrix
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 2 — Super Admin

Login: superadmin@qmis.edu / superadmin123

Modules in sidebar: 43

2.1 Dashboard

2.1.1 Dashboard

- **Path:** /super-admin/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

2.2 Attendance

2.2.1 Student Attendance

- **Path:** /super-admin/attendance/students/list
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/students|/admissions — Campus student SoT; Parent/Student portals
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

2.2.2 Employee Attendance

- **Path:** /super-admin/attendance/employees
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

2.3 User Database

2.3.1 Students List

- **Path:** /super-admin/user-database/students
- **Do:** Open **Students List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/students|/admissions — Campus student SoT; Parent/Student portals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/students|/admissions
- **Pass:** List/detail still correct after F5; no console/API 500.

2.3.2 Employee List

- **Path:** /super-admin/user-database/employees
- **Do:** Open **Employee List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

2.4 User Creation

2.4.1 User Creation

- **Path:** /super-admin/user-creation
- **Do:** Open **User Creation** (/super-admin/user-creation). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students
- **Pass:** List/detail still correct after F5; no console/API 500.

2.5 Gate Pass

2.5.1 Gate Pass

- **Path:** /super-admin/gate-pass
- **Do:** Open **Gate Pass**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Gate Keeper lists; PRM Front Office collections
- **Pass:** List/detail still correct after F5; no console/API 500.

2.6 Transport Overview

2.6.1 Transport Overview

- **Path:** /super-admin/transport-overview
- **Do:** Open **Transport Overview** (/super-admin/transport-overview). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.7 Finance

2.7.1 Finance Overview

- **Path:** /super-admin/finance/overview
- **Do:** Open **Finance Overview** (/super-admin/finance/overview). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

2.7.2 Fees Management

- **Path:** /super-admin/finance/fees
- **Do:** Open **Fees Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

2.7.3 Collections

- **Path:** /super-admin/finance/collections
- **Do:** Open **Collections**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*

- **Pass:** List/detail still correct after F5; no console/API 500.

2.7.4 Wallet Management

- **Path:** /super-admin/finance/wallets
- **Do:** Open **Wallet Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

2.7.5 Transport Finance

- **Path:** /super-admin/finance/transport
- **Do:** Open **Transport Finance** (/super-admin/finance/transport). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.7.6 Accounting

- **Path:** /super-admin/finance/accounting
- **Do:** Open **Accounting** (/super-admin/finance/accounting). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

2.7.7 Reports & Analytics

- **Path:** /super-admin/finance/reports
- **Do:** Open **Reports & Analytics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

2.8 Audit Reports

2.8.1 Overview

- **Path:** /super-admin/audit-reports/overview
- **Do:** Open **Overview** (/super-admin/audit-reports/overview). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

2.8.2 Compliance Score

- **Path:** /super-admin/audit-reports/compliance
- **Do:** Open **Compliance Score** (/super-admin/audit-reports/compliance). Exercise primary action (view/list/add). Refresh and confirm data behaviour.

- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

2.8.3 Department Ranking

- **Path:** /super-admin/audit-reports/department-ranking
- **Do:** Open **Department Ranking** (/super-admin/audit-reports/department-ranking). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

2.8.4 Pending Findings

- **Path:** /super-admin/audit-reports/pending-findings
- **Do:** Open **Pending Findings**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

2.8.5 Critical Findings

- **Path:** /super-admin/audit-reports/critical-findings
- **Do:** Open **Critical Findings**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

2.8.6 Risk Dashboard

- **Path:** /super-admin/audit-reports/risk-dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

2.9 Star Ratings

2.9.1 Star of the Month

- **Path:** /super-admin/star-ratings/star-of-month
- **Do:** Open **Star of the Month** (/super-admin/star-ratings/star-of-month). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

2.9.2 Star of the Year

- **Path:** /super-admin/star-ratings/star-of-year
- **Do:** Open **Star of the Year** (/super-admin/star-ratings/star-of-year). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

2.10 Task Management

2.10.1 Task Management

- **Path:** /super-admin/task-management/assign-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

2.11 Leave Request

2.11.1 Leave Request

- **Path:** /super-admin/leave-request/received
- **Do:** File a leave/permission request; note ID; check Received on approver login.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

2.12 Approvals

2.12.1 Approvals

- **Path:** /super-admin/approvals
- **Do:** Open **Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

2.13 Activities

2.13.1 Cultural

- **Path:** /super-admin/activities/cultural-list
- **Do:** Open **Cultural** (/super-admin/activities/cultural-list). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.13.2 Sports

- **Path:** /super-admin/activities/sports-list

- **Do:** Open **Sports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.13.3 Competitions

- **Path:** /super-admin/activities/competitions-list
- **Do:** Open **Competitions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.14 Announcement

2.14.1 Announcement

- **Path:** /super-admin/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

2.15 Communication

2.15.1 Communication

- **Path:** /super-admin/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

2.16 Calendar

2.16.1 Calendar

- **Path:** /super-admin/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

2.17 Notifications

2.17.1 Notifications

- **Path:** /super-admin/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

2.18 Escalation Management

2.18.1 Escalation Management

- **Path:** /super-admin/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

2.19 Activity Logs

2.19.1 Login History

- **Path:** /super-admin/activity-logs/login-history
- **Do:** Open **Login History** (/super-admin/activity-logs/login-history). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.19.2 Data Changes

- **Path:** /super-admin/activity-logs/data-changes
- **Do:** Open **Data Changes**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.19.3 Deleted Records

- **Path:** /super-admin/activity-logs/deleted-records
- **Do:** Open **Deleted Records**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.19.4 Approval Actions

- **Path:** /super-admin/activity-logs/approval-actions
- **Do:** Open **Approval Actions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.19.5 Audit Logs

- **Path:** /super-admin/activity-logs/audit-logs
- **Do:** Open **Audit Logs**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role

- **Pass:** List/detail still correct after F5; no console/API 500.

2.19.6 Failed Logins

- **Path:** /super-admin/activity-logs/failed-logins
- **Do:** Open **Failed Logins**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.19.7 Demo Activity

- **Path:** /super-admin/activity-logs/demo-activity
- **Do:** Open **Demo Activity** (/super-admin/activity-logs/demo-activity). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

2.20 Re-entry Approvals

2.20.1 Re-entry Approvals

- **Path:** /super-admin/reentry
- **Do:** Open **Re-entry Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

2.21 Module Health

2.21.1 Module Health

- **Path:** /super-admin/module-health
- **Do:** Open **Module Health** (/super-admin/module-health). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 3 — HR

Login: hr@qmis.edu / hr12345

Modules in sidebar: 36

3.1 Dashboard

3.1.1 Dashboard

- **Path:** /hr/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.2 Staff Management

3.2.1 Employees

- **Path:** /hr/employee-management/employees
- **Do:** Open **Employees**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.2.2 Employee Documents

- **Path:** /hr/employee-management/documents
- **Do:** Open **Employee Documents**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.3 Recruitment

3.3.1 Job Openings

- **Path:** /hr/recruitment/job-openings
- **Do:** Open **Job Openings**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

3.3.2 Candidates

- **Path:** /hr/recruitment/candidates
- **Do:** Open **Candidates**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

3.3.3 Interviews

- **Path:** /hr/recruitment/interviews
- **Do:** Open **Interviews**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

3.3.4 Interview Feedback

- **Path:** /hr/recruitment/interview-feedback
- **Do:** Open **Interview Feedback** (/hr/recruitment/interview-feedback). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

3.4 Onboarding

3.4.1 Offer Letters

- **Path:** /hr/onboarding/offers
- **Do:** Open **Offer Letters**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.4.2 Appointment Letters

- **Path:** /hr/onboarding/appointment
- **Do:** Open **Appointment Letters**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.4.3 New Joiner Details

- **Path:** /hr/onboarding/joiners
- **Do:** Open **New Joiner Details**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.4.4 New Joinee Observation

- **Path:** /hr/onboarding/observations
- **Do:** Open **New Joinee Observation** (/hr/onboarding/observations). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

3.4.5 Shadow Mentor

- **Path:** /hr/onboarding/shadow-mentor
- **Do:** Open **Shadow Mentor** (/hr/onboarding/shadow-mentor). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.4.6 Checklist

- **Path:** /hr/onboarding-checklist
- **Do:** Open **Checklist**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.5 Attendance & Leave

3.5.1 Attendance

- **Path:** /hr/attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

3.5.2 Leave Management

- **Path:** /hr/leave-management
- **Do:** Open **Leave Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

3.5.3 Staff Leave Requests

- **Path:** /hr/leave-request/received
- **Do:** Open **Staff Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

3.5.4 Leave Policies

- **Path:** /hr/leave-management/policies
- **Do:** Open **Leave Policies**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

3.6 Training

3.6.1 Training Schedule

- **Path:** /hr/training
- **Do:** Open **Training Schedule** (/hr/training). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.6.2 Training Records

- **Path:** /hr/training-records
- **Do:** Open **Training Records**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.6.3 Training Feedback

- **Path:** /hr/training-feedback
- **Do:** Open **Training Feedback** (/hr/training-feedback). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.7 Performance

3.7.1 Performance Review

- **Path:** /hr/performance-review
- **Do:** Open **Performance Review** (/hr/performance-review). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.7.2 Performance Comparison

- **Path:** /hr/performance-comparison
- **Do:** Open **Performance Comparison** (/hr/performance-comparison). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.7.3 BSC / Audit Score

- **Path:** /hr/performance-bsc
- **Do:** Open **BSC / Audit Score** (/hr/performance-bsc). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.7.4 Increment / Promotion

- **Path:** /hr/performance-increment
- **Do:** Open **Increment / Promotion** (/hr/performance-increment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.8 Payroll & Benefits

3.8.1 Salary Statement

- **Path:** /hr/payroll/salary-statement
- **Do:** Open **Salary Statement** (/hr/payroll/salary-statement). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.8.2 Payslip

- **Path:** /hr/payroll/payslip
- **Do:** Open **Payslip** (/hr/payroll/payslip). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.8.3 Salary Advance / Loan

- **Path:** /hr/payroll/salary-advance
- **Do:** Open **Salary Advance / Loan** (/hr/payroll/salary-advance). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.8.4 CTC

- **Path:** /hr/payroll/ctc
- **Do:** Open **CTC** (/hr/payroll/ctc). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.8.5 Referral Bonus

- **Path:** /hr/payroll/referral-bonus
- **Do:** Open **Referral Bonus**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.8.6 Employee Child Concession

- **Path:** /hr/payroll/child-concession
- **Do:** Open **Employee Child Concession** (/hr/payroll/child-concession). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.8.7 Claim Compensation

- **Path:** /hr/payroll/claim-compensation
- **Do:** Open **Claim Compensation** (/hr/payroll/claim-compensation). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.9 Disciplinary Action

3.9.1 Disciplinary Action

- **Path:** /hr/disciplinary
- **Do:** Open **Disciplinary Action** (/hr/disciplinary). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.10 Exit Formalities

3.10.1 Exit Formalities

- **Path:** /hr/exit
- **Do:** Open **Exit Formalities**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.11 Reports

3.11.1 Reports

- **Path:** /hr/reports
- **Do:** Open **Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

3.12 Notifications / Announcements

3.12.1 Notifications

- **Path:** /hr/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.

- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

3.12.2 Announcements

- **Path:** /hr/announcements
- **Do:** Open **Announcements**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 4 — Account Head

Login: accounthead@qmis.edu / accounts123

Modules in sidebar: 22

4.1 Dashboard

4.1.1 Dashboard

- **Path:** /account-head/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.2 Fees Management

4.2.1 Fees Management

- **Path:** /account-head/fees-management
- **Do:** Open **Fees Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

4.3 Collections

4.3.1 Collections

- **Path:** /account-head/collections
- **Do:** Open **Collections**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

4.4 Wallet Management

4.4.1 Wallet Management

- **Path:** /account-head/wallet-management
- **Do:** Open **Wallet Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.5 Transport Finance

4.5.1 Transport Finance

- **Path:** /account-head/transport-finance
- **Do:** Open **Transport Finance** (/account-head/transport-finance). Exercise primary action (view/list/add). Refresh and confirm data behaviour.

- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6 Accounting

4.6.1 Day Book

- **Path:** /account-head/accounting/day-book
- **Do:** Open **Day Book** (/account-head/accounting/day-book). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.2 Online Book

- **Path:** /account-head/accounting/online-book
- **Do:** Open **Online Book** (/account-head/accounting/online-book). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.3 General Ledger

- **Path:** /account-head/accounting/general-ledger
- **Do:** Open **General Ledger** (/account-head/accounting/general-ledger). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.4 Cash Book

- **Path:** /account-head/accounting/cash-book
- **Do:** Open **Cash Book** (/account-head/accounting/cash-book). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.5 Bank Book

- **Path:** /account-head/accounting/bank-book
- **Do:** Open **Bank Book** (/account-head/accounting/bank-book). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.6 Chart of Accounts

- **Path:** /account-head/accounting/chart-of-accounts
- **Do:** Open **Chart of Accounts**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.7 Journal Vouchers

- **Path:** /account-head/accounting/journal-vouchers
- **Do:** Open **Journal Vouchers**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.8 Bank Reconciliation

- **Path:** /account-head/accounting/bank-reconciliation
- **Do:** Open **Bank Reconciliation** (/account-head/accounting/bank-reconciliation). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.6.9 Financial Statements

- **Path:** /account-head/accounting/financial-statements
- **Do:** Open **Financial Statements**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.7 Approvals

4.7.1 Approvals

- **Path:** /account-head/approvals
- **Do:** Open **Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.8 Reports & Analytics

4.8.1 Reports & Analytics

- **Path:** /account-head/reports-analytics

- **Do:** Open **Reports & Analytics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.9 Task Management

4.9.1 Task Management

- **Path:** /account-head/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

4.10 Settings

4.10.1 Settings

- **Path:** /account-head/settings
- **Do:** Open **Settings**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.11 Payment Vouchers

4.11.1 Payment Vouchers

- **Path:** /account-head/payment-vouchers
- **Do:** Open **Payment Vouchers**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

4.12 Book Fee

4.12.1 Book Fee

- **Path:** /account-head/book-fee
- **Do:** Open **Book Fee** (/account-head/book-fee). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

4.13 Annual Budget

4.13.1 Annual Budget

- **Path:** /account-head/annual-budget

- **Do:** Open **Annual Budget** (/account-head/annual-budget). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/finance/* — Fee ledger; Parent payment views; approvals
- **Reflects to:** Same role after refresh; related roles that share /api/v1/finance/*
- **Pass:** List/detail still correct after F5; no console/API 500.

4.14 MFP Report

4.14.1 MFP Report

- **Path:** /account-head/mfp
- **Do:** Open **MFP Report** (/account-head/mfp). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 5 — PRM (Front Office)

Login: prm@qmis.edu / prm12345

Modules in sidebar: 21

5.1 Admission Enquiry

5.1.1 Admission Enquiry

- **Path:** /front-office/admission-enquiry
- **Do:** Open **Admission Enquiry** (/front-office/admission-enquiry). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Student Database; Parent login; class lists
- **Pass:** List/detail still correct after F5; no console/API 500.

5.2 Admission List

5.2.1 Admission List

- **Path:** /front-office/admission-list
- **Do:** Open **Admission List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Student Database; Parent login; class lists
- **Pass:** List/detail still correct after F5; no console/API 500.

5.3 User Database

5.3.1 Student Database

- **Path:** /front-office/student-management
- **Do:** Open **Student Database** (/front-office/student-management). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

5.3.2 Teacher Database

- **Path:** /front-office/teacher-management
- **Do:** Open **Teacher Database** (/front-office/teacher-management). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

5.4 Student Transfer

5.4.1 Student Transfer

- **Path:** /front-office/student-transfer
- **Do:** Open **Student Transfer** (/front-office/student-transfer). Exercise primary action (view/list/add). Refresh and confirm data behaviour.

- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admissions or /api/v1/front-office/{collection}
- **Pass:** List/detail still correct after F5; no console/API 500.

5.5 Student Re-Enrollment

5.5.1 Student Re-Enrollment

- **Path:** /front-office/student-re-enrollment
- **Do:** Open **Student Re-Enrollment** (/front-office/student-re-enrollment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Student Database; Parent login; class lists
- **Pass:** List/detail still correct after F5; no console/API 500.

5.6 TC Request

5.6.1 TC Request

- **Path:** /front-office/tc-request
- **Do:** Open **TC Request** (/front-office/tc-request). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/tc-requests — Parent/Student file → PRM forward → Admin decide
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tc-requests
- **Pass:** List/detail still correct after F5; no console/API 500.

5.7 Gate Pass

5.7.1 Student Gate Pass

- **Path:** /front-office/gate-pass-list
- **Do:** Open **Student Gate Pass**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Gate Keeper lists; PRM Front Office collections
- **Pass:** List/detail still correct after F5; no console/API 500.

5.7.2 Hostel Gate Pass

- **Path:** /front-office/hostel-gate-pass-list
- **Do:** Open **Hostel Gate Pass**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Gate Keeper lists; PRM Front Office collections
- **Pass:** List/detail still correct after F5; no console/API 500.

5.7.3 Material Gate Pass

- **Path:** /front-office/material-gate-pass-list
- **Do:** Open **Material Gate Pass**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Gate Keeper lists; PRM Front Office collections
- **Pass:** List/detail still correct after F5; no console/API 500.

5.7.4 Goods Received Pass

- **Path:** /front-office/goods-received-pass-list
- **Do:** Open **Goods Received Pass**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admissions or /api/v1/front-office/{collection}
- **Pass:** List/detail still correct after F5; no console/API 500.

5.8 Task Management

5.8.1 Assign Tasks

- **Path:** /front-office/task-management/assign-tasks
- **Do:** Open **Assign Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

5.8.2 My Tasks

- **Path:** /front-office/task-management/my-tasks
- **Do:** Open **My Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

5.9 Leave Request

5.9.1 My Leave Requests

- **Path:** /front-office/leave-request/my-requests
- **Do:** Open **My Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

5.9.2 Received Leave Requests

- **Path:** /front-office/leave-request/received
- **Do:** Open **Received Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

5.10 My Attendance

5.10.1 My Attendance

- **Path:** /front-office/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)

- **Pass:** List/detail still correct after F5; no console/API 500.

5.11 Announcement

5.11.1 Announcement

- **Path:** /front-office/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

5.12 Communication

5.12.1 Communication

- **Path:** /front-office/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

5.13 Calendar

5.13.1 Calendar

- **Path:** /front-office/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

5.14 Notifications

5.14.1 Notifications

- **Path:** /front-office/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

5.15 Escalation Management

5.15.1 Escalation Management

- **Path:** /front-office/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 6 — Principal

Login: principal@qmis.edu / principal123

Modules in sidebar: 21

6.1 Task Management

6.1.1 Assign Tasks

- **Path:** /principal/task-management/assign-tasks
- **Do:** Open **Assign Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

6.1.2 My Tasks

- **Path:** /principal/task-management/my-tasks
- **Do:** Open **My Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

6.2 Star Ratings

6.2.1 Star of the Month

- **Path:** /principal/star-ratings/star-of-month
- **Do:** Open **Star of the Month** (/principal/star-ratings/star-of-month). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

6.2.2 Star of the Year

- **Path:** /principal/star-ratings/star-of-year
- **Do:** Open **Star of the Year** (/principal/star-ratings/star-of-year). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

6.3 Examination Timetable

6.3.1 Examination Timetable

- **Path:** /principal/examination-timetable
- **Do:** Open **Examination Timetable** (/principal/examination-timetable). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/academics/{collection} — Principal write; Teacher change-requests; Student/Parent read published
- **Reflects to:** Teacher/Coordinator/Student/Parent timetable views
- **Pass:** List/detail still correct after F5; no console/API 500.

6.4 Class Timetable

6.4.1 Class Timetable

- **Path:** /principal/class-timetable
- **Do:** Open **Class Timetable** (/principal/class-timetable). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/academics/{collection} — Principal write; Teacher change-requests; Student/Parent read published
- **Reflects to:** Teacher/Coordinator/Student/Parent timetable views
- **Pass:** List/detail still correct after F5; no console/API 500.

6.5 User Database

6.5.1 Student Database

- **Path:** /principal/user-management/student-database
- **Do:** Open **Student Database** (/principal/user-management/student-database). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students
- **Pass:** List/detail still correct after F5; no console/API 500.

6.5.2 Employee Database

- **Path:** /principal/user-management/employee-database
- **Do:** Open **Employee Database** (/principal/user-management/employee-database). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

6.6 Teacher Allocation

6.6.1 Teacher Allocation

- **Path:** /principal/academic/teacher-allocation
- **Do:** Open **Teacher Allocation** (/principal/academic/teacher-allocation). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/academics/{collection} — Principal write; Teacher change-requests; Student/Parent read published
- **Reflects to:** Teacher/Coordinator/Student/Parent timetable views
- **Pass:** List/detail still correct after F5; no console/API 500.

6.7 Student Allocation

6.7.1 Student Allocation

- **Path:** /principal/student-allocation-approval
- **Do:** Open **Student Allocation** (/principal/student-allocation-approval). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Teacher/Coordinator/Student/Parent timetable views
- **Pass:** List/detail still correct after F5; no console/API 500.

6.8 Extended Class

6.8.1 Extended Class

- **Path:** /principal/class/extra-class
- **Do:** Open **Extended Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

6.9 Home Fun

6.9.1 Student Home Fun

- **Path:** /principal/lms/student-lms
- **Do:** Open **Student Home Fun** (/principal/lms/student-lms). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

6.9.2 Teacher Home Fun

- **Path:** /principal/lms/teacher-lms
- **Do:** Open **Teacher Home Fun** (/principal/lms/teacher-lms). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

6.10 Leave Request

6.10.1 My Leave Requests

- **Path:** /principal/leave-request/my-requests
- **Do:** Open **My Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

6.10.2 Received Leave Requests

- **Path:** /principal/leave-request/received
- **Do:** Open **Received Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

6.11 My Attendance

6.11.1 My Attendance

- **Path:** /principal/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)
- **Pass:** List/detail still correct after F5; no console/API 500.

6.12 Announcement

6.12.1 Announcement

- **Path:** /principal/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

6.13 Communication

6.13.1 Communication

- **Path:** /principal/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

6.14 Calendar

6.14.1 Calendar

- **Path:** /principal/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

6.15 Notifications

6.15.1 Notifications

- **Path:** /principal/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

6.16 Escalation Management

6.16.1 Escalation Management

- **Path:** /principal/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications

- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 7 — Director

Login: director@qmis.edu / director123

Modules in sidebar: 24

7.1 Task Management

7.1.1 Assign Tasks

- **Path:** /director/task-management/assign-tasks
- **Do:** Open **Assign Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

7.1.2 My Tasks

- **Path:** /director/task-management/my-tasks
- **Do:** Open **My Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

7.2 Star Ratings

7.2.1 Star of the Month

- **Path:** /director/star-ratings/star-of-month
- **Do:** Open **Star of the Month** (/director/star-ratings/star-of-month). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

7.2.2 Star of the Year

- **Path:** /director/star-ratings/star-of-year
- **Do:** Open **Star of the Year** (/director/star-ratings/star-of-year). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

7.3 Timetable Approvals

7.3.1 Exam Timetable

- **Path:** /director/examination-approvals
- **Do:** Open **Exam Timetable** (/director/examination-approvals). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Same role after refresh; related roles that share /api/v1/exams|marks|results
- **Pass:** List/detail still correct after F5; no console/API 500.

7.4 User Database

7.4.1 Student Database

- **Path:** /director/user-management/student-database
- **Do:** Open **Student Database** (/director/user-management/student-database). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students
- **Pass:** List/detail still correct after F5; no console/API 500.

7.4.2 Employee Database

- **Path:** /director/user-management/employee-database
- **Do:** Open **Employee Database** (/director/user-management/employee-database). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

7.5 Activities

7.5.1 Cultural

- **Path:** /director/activities/cultural-list
- **Do:** Open **Cultural** (/director/activities/cultural-list). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

7.5.2 Sports

- **Path:** /director/activities/sports-list
- **Do:** Open **Sports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

7.5.3 Chess Result

- **Path:** /director/activities/chess-result
- **Do:** Open **Chess Result** (/director/activities/chess-result). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Parent/Student Exam Result after publish
- **Pass:** List/detail still correct after F5; no console/API 500.

7.5.4 Competitions

- **Path:** /director/activities/competitions-list
- **Do:** Open **Competitions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

7.6 Lesson Plan Approval

7.6.1 Lesson Plan Approval

- **Path:** /director/lesson-plan-approval
- **Do:** Open **Lesson Plan Approval** (/director/lesson-plan-approval). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/lesson-plans — Teacher submit → Coordinator/Principal/Director approve
- **Reflects to:** Coordinator/Principal/Director approval queues
- **Pass:** List/detail still correct after F5; no console/API 500.

7.7 Mark Entry Approval

7.7.1 Mark Entry Approval

- **Path:** /director/mark-entry-approval
- **Do:** Open **Mark Entry Approval** (/director/mark-entry-approval). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Parent/Student Exam Result after publish
- **Pass:** List/detail still correct after F5; no console/API 500.

7.8 Home Fun

7.8.1 Student Home Fun

- **Path:** /director/lms/student-lms
- **Do:** Open **Student Home Fun** (/director/lms/student-lms). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

7.8.2 Teacher Home Fun

- **Path:** /director/lms/teacher-lms
- **Do:** Open **Teacher Home Fun** (/director/lms/teacher-lms). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

7.9 Request Approvals

7.9.1 Request Approvals

- **Path:** /director/request-approvals
- **Do:** Open **Request Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

7.10 Leave Request

7.10.1 My Leave Requests

- **Path:** /director/leave-request/my-requests
- **Do:** Open **My Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

7.10.2 Received Leave Requests

- **Path:** /director/leave-request/received
- **Do:** Open **Received Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

7.11 My Attendance

7.11.1 My Attendance

- **Path:** /director/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)
- **Pass:** List/detail still correct after F5; no console/API 500.

7.12 Announcement

7.12.1 Announcement

- **Path:** /director/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

7.13 Communication

7.13.1 Communication

- **Path:** /director/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

7.14 Calendar

7.14.1 Calendar

- **Path:** /director/academic-calendar

- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

7.15 Notifications

7.15.1 Notifications

- **Path:** /director/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

7.16 Escalation Management

7.16.1 Escalation Management

- **Path:** /director/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 8 — Coordinator

Login: coordinator@qmis.edu / coord123

Modules in sidebar: 30

8.1 Dashboard

8.1.1 Dashboard

- **Path:** /coordinator/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.2 Attendance

8.2.1 Class Attendance

- **Path:** /coordinator/attendance/class-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/attendance — Class attendance; Parent/Student can read published
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

8.3 Class

8.3.1 Class Timetable

- **Path:** /coordinator/class/class-routine
- **Do:** Open **Class Timetable** (/coordinator/class/class-routine). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.3.2 Lesson Plan

- **Path:** /coordinator/lesson-plan/my-lesson-plan
- **Do:** Open **Lesson Plan** (/coordinator/lesson-plan/my-lesson-plan). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/lesson-plans — Teacher submit → Coordinator/Principal/Director approve
- **Reflects to:** Coordinator/Principal/Director approval queues
- **Pass:** List/detail still correct after F5; no console/API 500.

8.3.3 Lesson Plan Approval

- **Path:** /coordinator/lesson-plan-approval
- **Do:** Open **Lesson Plan Approval** (/coordinator/lesson-plan-approval). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/lesson-plans — Teacher submit → Coordinator/Principal/Director approve
- **Reflects to:** Coordinator/Principal/Director approval queues
- **Pass:** List/detail still correct after F5; no console/API 500.

8.3.4 Extended Class

- **Path:** /coordinator/class/extra-class
- **Do:** Open **Extended Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.3.5 Online Class

- **Path:** /coordinator/class/online-class
- **Do:** Open **Online Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.4 Task Management

8.4.1 Task Management

- **Path:** /coordinator/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5 Student

8.5.1 Students List

- **Path:** /coordinator/user-management/students-list
- **Do:** Open **Students List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.2 Parents List

- **Path:** /coordinator/user-management/parents-list
- **Do:** Open **Parents List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.3 CA / RCT

- **Path:** /coordinator/unit-tests
- **Do:** Open **CA / RCT** (/coordinator/unit-tests). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen

- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.4 Exam

- **Path:** /coordinator/student-evaluation/exam
- **Do:** Open **Exam** (/coordinator/student-evaluation/exam). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Same role after refresh; related roles that share /api/v1/exams|marks|results
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.5 Question Bank

- **Path:** /coordinator/student-deliverables/sample-questions
- **Do:** Open **Question Bank** (/coordinator/student-deliverables/sample-questions). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.6 Mark Entry

- **Path:** /coordinator/student-evaluation/mark-entry
- **Do:** Open **Mark Entry** (/coordinator/student-evaluation/mark-entry). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Parent/Student Exam Result after publish
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.7 Performance Analytics

- **Path:** /coordinator/student-evaluation/analytics
- **Do:** Open **Performance Analytics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.8 Biometric Attendance

- **Path:** /coordinator/attendance/biometric
- **Do:** Click Sync Biometric Data; refresh; confirm rows remain. Teacher: try one override.
- **Stored in:** GET/PUT /api/v1/ops/biometric-attendance — Teacher overrides; Coordinator views; punches simulated until eSSL import
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/ops/biometric-attendance
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.9 Home Fun

- **Path:** /coordinator/student-deliverables/home-fun
- **Do:** Open **Home Fun** (/coordinator/student-deliverables/home-fun). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps

- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.10 Study Materials

- **Path:** /coordinator/student-deliverables/study-materials
- **Do:** Open **Study Materials**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.11 Star of the Month

- **Path:** /coordinator/star-ratings/som-matrix
- **Do:** Open **Star of the Month** (/coordinator/star-ratings/som-matrix). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.5.12 Star of the Year

- **Path:** /coordinator/star-ratings/star-of-year
- **Do:** Open **Star of the Year** (/coordinator/star-ratings/star-of-year). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.6 My Profile

8.6.1 My Profile

- **Path:** /coordinator/my-profile
- **Do:** Open **My Profile** (/coordinator/my-profile). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

8.6.2 My Attendance

- **Path:** /coordinator/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)
- **Pass:** List/detail still correct after F5; no console/API 500.

8.6.3 My Leave Requests

- **Path:** /coordinator/leave-request/my-requests

- **Do:** Open **My Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

8.6.4 Received Leave Requests

- **Path:** /coordinator/leave-request/received
- **Do:** Open **Received Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

8.6.5 Books Borrowed

- **Path:** /coordinator/library/books-borrowed
- **Do:** Open **Books Borrowed** (/coordinator/library/books-borrowed). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/library-* — Librarian write; Parent/Student borrowed books if wired
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/library-*
- **Pass:** List/detail still correct after F5; no console/API 500.

8.7 Announcement

8.7.1 Announcement

- **Path:** /coordinator/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

8.8 Communication

8.8.1 Communication

- **Path:** /coordinator/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

8.9 Calendar

8.9.1 Calendar

- **Path:** /coordinator/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

8.10 Notifications

8.10.1 Notifications

- **Path:** /coordinator/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

8.11 Escalation Management

8.11.1 Escalation Management

- **Path:** /coordinator/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 9 — Teacher

Login: teacher@qmis.edu / teacher123

Modules in sidebar: 30

9.1 Dashboard

9.1.1 Dashboard

- **Path:** /teacher/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.2 Attendance

9.2.1 Class Attendance

- **Path:** /teacher/attendance/class-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/attendance — Class attendance; Parent/Student can read published
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

9.2.2 Biometric Attendance

- **Path:** /teacher/attendance/biometric
- **Do:** Click Sync Biometric Data; refresh; confirm rows remain. Teacher: try one override.
- **Stored in:** GET/PUT /api/v1/ops/biometric-attendance — Teacher overrides; Coordinator views; punches simulated until eSSL import
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/ops/biometric-attendance
- **Pass:** List/detail still correct after F5; no console/API 500.

9.3 Class

9.3.1 Class Timetable

- **Path:** /teacher/class/class-routine
- **Do:** Open **Class Timetable** (/teacher/class/class-routine). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.3.2 Lesson Plan

- **Path:** /teacher/lesson-plan/my-lesson-plan
- **Do:** Open **Lesson Plan** (/teacher/lesson-plan/my-lesson-plan). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/lesson-plans — Teacher submit → Coordinator/Principal/Director approve
- **Reflects to:** Coordinator/Principal/Director approval queues
- **Pass:** List/detail still correct after F5; no console/API 500.

9.3.3 Lesson Plan Approval

- **Path:** /teacher/lesson-plan-approval
- **Do:** Open **Lesson Plan Approval** (/teacher/lesson-plan-approval). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/lesson-plans — Teacher submit → Coordinator/Principal/Director approve
- **Reflects to:** Coordinator/Principal/Director approval queues
- **Pass:** List/detail still correct after F5; no console/API 500.

9.3.4 Extended Class

- **Path:** /teacher/class/extra-class
- **Do:** Open **Extended Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.3.5 Online Class

- **Path:** /teacher/class/online-class
- **Do:** Open **Online Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.4 Task Management

9.4.1 Task Management

- **Path:** /teacher/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5 Student

9.5.1 Students List

- **Path:** /teacher/user-management/students-list
- **Do:** Open **Students List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.2 Parents List

- **Path:** /teacher/user-management/parents-list
- **Do:** Open **Parents List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admin users / students — Creates logins used by every later role
- **Reflects to:** Same role after refresh; related roles that share /api/v1/admin users / students

- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.3 CA / RCT

- **Path:** /teacher/unit-tests
- **Do:** Open **CA / RCT** (/teacher/unit-tests). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.4 Exam

- **Path:** /teacher/student-evaluation/exam
- **Do:** Open **Exam** (/teacher/student-evaluation/exam). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Same role after refresh; related roles that share /api/v1/exams|marks|results
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.5 Question Bank

- **Path:** /teacher/student-deliverables/sample-questions
- **Do:** Open **Question Bank** (/teacher/student-deliverables/sample-questions). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.6 Mark Entry

- **Path:** /teacher/student-evaluation/mark-entry
- **Do:** Open **Mark Entry** (/teacher/student-evaluation/mark-entry). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Parent/Student Exam Result after publish
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.7 Performance Analytics

- **Path:** /teacher/student-evaluation/analytics
- **Do:** Open **Performance Analytics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.8 Home Fun

- **Path:** /teacher/student-deliverables/home-fun
- **Do:** Open **Home Fun** (/teacher/student-deliverables/home-fun). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps

- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.9 Study Materials

- **Path:** /teacher/student-deliverables/study-materials
- **Do:** Open **Study Materials**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.10 Star of the Month

- **Path:** /teacher/star-ratings/som-matrix
- **Do:** Open **Star of the Month** (/teacher/star-ratings/som-matrix). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.5.11 Star of the Year

- **Path:** /teacher/star-ratings/star-of-year
- **Do:** Open **Star of the Year** (/teacher/star-ratings/star-of-year). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.6 My Profile

9.6.1 My Profile

- **Path:** /teacher/my-profile
- **Do:** Open **My Profile** (/teacher/my-profile). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

9.6.2 My Attendance

- **Path:** /teacher/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)
- **Pass:** List/detail still correct after F5; no console/API 500.

9.6.3 My Leave Requests

- **Path:** /teacher/leave-request/my-requests

- **Do:** Open **My Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

9.6.4 Received Leave Requests

- **Path:** /teacher/leave-request/received
- **Do:** Open **Received Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

9.6.5 Books Borrowed

- **Path:** /teacher/library/books-borrowed
- **Do:** Open **Books Borrowed** (/teacher/library/books-borrowed). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/library-* — Librarian write; Parent/Student borrowed books if wired
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/library-*
- **Pass:** List/detail still correct after F5; no console/API 500.

9.7 Announcement

9.7.1 Announcement

- **Path:** /teacher/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

9.8 Communication

9.8.1 Communication

- **Path:** /teacher/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

9.9 Calendar

9.9.1 Calendar

- **Path:** /teacher/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

9.10 Notifications

9.10.1 Notifications

- **Path:** /teacher/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

9.11 Escalation Management

9.11.1 Escalation Management

- **Path:** /teacher/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 10 — Parent

Login: (from Admission enroll) / (set at enroll)

Modules in sidebar: 23

10.1 Dashboard

10.1.1 Dashboard

- **Path:** /parent/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.2 Class

10.2.1 Online Class

- **Path:** /parent/class/online-class
- **Do:** Open **Online Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.2.2 Extended Class

- **Path:** /parent/class/extra-class
- **Do:** Open **Extended Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.2.3 Timetable

- **Path:** /parent/class/timetable-list
- **Do:** Open **Timetable** (/parent/class/timetable-list). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/academics/{collection} — Principal write; Teacher change-requests; Student/Parent read published
- **Reflects to:** Teacher/Coordinator/Student/Parent timetable views
- **Pass:** List/detail still correct after F5; no console/API 500.

10.2.4 Attendance List

- **Path:** /parent/class/attendance-list
- **Do:** Open **Attendance List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

10.3 Student Evaluation

10.3.1 Exam Result

- **Path:** /parent/student-evaluation/exam-result
- **Do:** Open **Exam Result** (/parent/student-evaluation/exam-result). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Parent/Student Exam Result after publish
- **Pass:** List/detail still correct after F5; no console/API 500.

10.3.2 Exam Schedule

- **Path:** /parent/student-evaluation/exam-schedule
- **Do:** Open **Exam Schedule** (/parent/student-evaluation/exam-schedule). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Same role after refresh; related roles that share /api/v1/exams|marks|results
- **Pass:** List/detail still correct after F5; no console/API 500.

10.3.3 Performance Analytics

- **Path:** /parent/student-evaluation/analytics
- **Do:** Open **Performance Analytics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.4 Student Deliverables

10.4.1 Home Fun

- **Path:** /parent/student-deliverables/home-fun
- **Do:** Open **Home Fun** (/parent/student-deliverables/home-fun). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.4.2 Study Materials

- **Path:** /parent/student-deliverables/study-materials
- **Do:** Open **Study Materials**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.4.3 Sample Questions

- **Path:** /parent/student-deliverables/sample-questions

- **Do:** Open **Sample Questions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.5 Library

10.5.1 Borrowed Books

- **Path:** /parent/library/borrowed-books-list
- **Do:** Open **Borrowed Books**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/library-* — Librarian write; Parent/Student borrowed books if wired
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/library-*
- **Pass:** List/detail still correct after F5; no console/API 500.

10.6 Transport

10.6.1 Track Bus

- **Path:** /parent/transport/track-bus
- **Do:** Open **Track Bus**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/protrack/devices|track — Live GPS if PROTRACK_* in backend/.env and ProTrack Open API allowed
- **Reflects to:** Parent Track Bus (parent panel); TM Track (admin panel)
- **Pass:** List/detail still correct after F5; no console/API 500.

10.7 Hostel Details

10.7.1 Hostel Details

- **Path:** /parent/hostel/hostel-details
- **Do:** Open **Hostel Details**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.8 Fees Payment

10.8.1 Academic Payment

- **Path:** /parent/payment/fees-payment
- **Do:** Open **Academic Payment** (/parent/payment/fees-payment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.8.2 Hostel Payment

- **Path:** /parent/payment/hostel-payment

- **Do:** Open **Hostel Payment** (/parent/payment/hostel-payment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

10.8.3 Transport Payment

- **Path:** /parent/payment/transport-payment
- **Do:** Open **Transport Payment** (/parent/payment/transport-payment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

10.9 Announcement

10.9.1 Announcement

- **Path:** /parent/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

10.10 TC Request

10.10.1 TC Request

- **Path:** /parent/tc-request
- **Do:** Open **TC Request** (/parent/tc-request). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/tc-requests — Parent/Student file → PRM forward → Admin decide
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tc-requests
- **Pass:** List/detail still correct after F5; no console/API 500.

10.11 Communication

10.11.1 Communication

- **Path:** /parent/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

10.12 Calendar

10.12.1 Calendar

- **Path:** /parent/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar

- **Pass:** List/detail still correct after F5; no console/API 500.

10.13 Notifications

10.13.1 Notifications

- **Path:** /parent/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

10.14 Escalation Management

10.14.1 Escalation Management

- **Path:** /parent/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 11 — Student

Login: (optional student login) / (if provisioned)

Modules in sidebar: 23

11.1 Class

11.1.1 Online Class

- **Path:** /student/class/online-class
- **Do:** Open **Online Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.1.2 Extended Class

- **Path:** /student/class/extra-class
- **Do:** Open **Extended Class**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.1.3 Timetable

- **Path:** /student/class/timetable-list
- **Do:** Open **Timetable** (/student/class/timetable-list). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/academics/{collection} — Principal write; Teacher change-requests; Student/Parent read published
- **Reflects to:** Teacher/Coordinator/Student/Parent timetable views
- **Pass:** List/detail still correct after F5; no console/API 500.

11.1.4 Attendance List

- **Path:** /student/class/attendance-list
- **Do:** Open **Attendance List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

11.2 Student Evaluation

11.2.1 Exam Result

- **Path:** /student/student-evaluation/exam-result
- **Do:** Open **Exam Result** (/student/student-evaluation/exam-result). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Parent/Student Exam Result after publish
- **Pass:** List/detail still correct after F5; no console/API 500.

11.2.2 Exam Schedule

- **Path:** /student/student-evaluation/exam-schedule
- **Do:** Open **Exam Schedule** (/student/student-evaluation/exam-schedule). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/exams|marks|results — Teacher enter → publish → Parent/Student results
- **Reflects to:** Same role after refresh; related roles that share /api/v1/exams|marks|results
- **Pass:** List/detail still correct after F5; no console/API 500.

11.3 Student Deliverables

11.3.1 Home Fun

- **Path:** /student/student-deliverables/home-fun
- **Do:** Open **Home Fun** (/student/student-deliverables/home-fun). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.3.2 Study Materials

- **Path:** /student/student-deliverables/study-materials
- **Do:** Open **Study Materials**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.3.3 Sample Questions

- **Path:** /student/student-deliverables/sample-questions
- **Do:** Open **Sample Questions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.4 Star Ratings

11.4.1 Star Ratings

- **Path:** /student/star-ratings/view-ratings
- **Do:** Open **Star Ratings**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.5 Library

11.5.1 Borrowed Books

- **Path:** /student/library/borrowed-books-list
- **Do:** Open **Borrowed Books**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/library-* — Librarian write; Parent/Student borrowed books if wired
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/library-*
- **Pass:** List/detail still correct after F5; no console/API 500.

11.6 Transport

11.6.1 Bus Route

- **Path:** /student/transport/bus-route-details
- **Do:** Open **Bus Route** (/student/transport/bus-route-details). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

11.6.2 Track Bus

- **Path:** /student/transport/track-bus
- **Do:** Open **Track Bus**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/protrack/devices|track — Live GPS if PROTRACK_* in backend/.env and ProTrack Open API allowed
- **Reflects to:** Parent Track Bus (parent panel); TM Track (admin panel)
- **Pass:** List/detail still correct after F5; no console/API 500.

11.7 Hostel Details

11.7.1 Hostel Details

- **Path:** /student/hostel/hostel-details
- **Do:** Open **Hostel Details**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.8 Payment Details

11.8.1 Fees Payment

- **Path:** /student/payment/fees-payment
- **Do:** Open **Fees Payment** (/student/payment/fees-payment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.8.2 Hostel Payment

- **Path:** /student/payment/hostel-payment
- **Do:** Open **Hostel Payment** (/student/payment/hostel-payment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.

- **Stored in:** Campus API or ops/audit collection for this screen — Refresh must keep row if SoT-backed; demo-only screens noted in Known gaps
- **Reflects to:** Same role after refresh; related roles that share Campus API or ops/audit collection for this screen
- **Pass:** List/detail still correct after F5; no console/API 500.

11.8.3 Transport Payment

- **Path:** /student/payment/transport-payment
- **Do:** Open **Transport Payment** (/student/payment/transport-payment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

11.9 Announcement

11.9.1 Announcement

- **Path:** /student/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

11.10 TC Request

11.10.1 TC Request

- **Path:** /student/tc-request
- **Do:** Open **TC Request** (/student/tc-request). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/tc-requests — Parent/Student file → PRM forward → Admin decide
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tc-requests
- **Pass:** List/detail still correct after F5; no console/API 500.

11.11 Communication

11.11.1 Communication

- **Path:** /student/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

11.12 Calendar

11.12.1 Calendar

- **Path:** /student/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

11.13 Notifications

11.13.1 Notifications

- **Path:** /student/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

11.14 Escalation Management

11.14.1 Escalation Management

- **Path:** /student/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 12 — Librarian

Login: (HR create staff login / seed if present) / —

Modules in sidebar: 10

12.1 Book Management

12.1.1 Book List

- **Path:** /librarian/book-management/book-list
- **Do:** Open **Book List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/library-* — Librarian write; Parent/Student borrowed books if wired
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/library-*
- **Pass:** List/detail still correct after F5; no console/API 500.

12.2 Issued Books

12.2.1 Issued Books

- **Path:** /librarian/issued-books/issued-book-list
- **Do:** Open **Issued Books**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/library-* — Librarian write; Parent/Student borrowed books if wired
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/library-*
- **Pass:** List/detail still correct after F5; no console/API 500.

12.3 Members

12.3.1 Members

- **Path:** /librarian/members/member-list
- **Do:** Open **Members**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/library-* — Librarian write; Parent/Student borrowed books if wired
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/library-*
- **Pass:** List/detail still correct after F5; no console/API 500.

12.4 Task Management

12.4.1 Task Management

- **Path:** /librarian/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

12.5 My Attendance

12.5.1 My Attendance

- **Path:** /librarian/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)
- **Pass:** List/detail still correct after F5; no console/API 500.

12.6 Announcement

12.6.1 Announcement

- **Path:** /librarian/announcement
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Every role in the audience sidebar Announcement list
- **Pass:** List/detail still correct after F5; no console/API 500.

12.7 Communication

12.7.1 Communication

- **Path:** /librarian/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

12.8 Calendar

12.8.1 Calendar

- **Path:** /librarian/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

12.9 Notifications

12.9.1 Notifications

- **Path:** /librarian/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

12.10 Escalation Management

12.10.1 Escalation Management

- **Path:** /librarian/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 13 — Gate Keeper

Login: (HR create staff login) / —

Modules in sidebar: 16

13.1 Hostel Gate Pass

13.1.1 Hostel Gate Pass

- **Path:** /gate-keeper/hostel-gate-pass
- **Do:** Open **Hostel Gate Pass**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Gate Keeper lists; PRM Front Office collections
- **Pass:** List/detail still correct after F5; no console/API 500.

13.2 My Duty

13.2.1 My Duty

- **Path:** /gate-keeper/my-duty
- **Do:** Open **My Duty** (/gate-keeper/my-duty). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

13.3 CSO / Gate Keeper

13.3.1 CSO / Gate Keeper

- **Path:** /gate-keeper/incidents
- **Do:** Open **CSO / Gate Keeper** (/gate-keeper/incidents). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

13.4 Gate Pass

13.4.1 Gate Pass

- **Path:** /gate-keeper/gate-pass-list
- **Do:** Open **Gate Pass**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/admissions or /api/v1/front-office/{collection} — PRM write; Principal/Director/Gate may read/write passes
- **Reflects to:** Gate Keeper lists; PRM Front Office collections
- **Pass:** List/detail still correct after F5; no console/API 500.

13.5 Task Management

13.5.1 Task Management

- **Path:** /gate-keeper/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks

- **Pass:** List/detail still correct after F5; no console/API 500.

13.6 Leave Request

13.6.1 Leave Request

- **Path:** /gate-keeper/leave-request/my-requests
- **Do:** File a leave/permission request; note ID; check Received on approver login.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

13.7 My Attendance

13.7.1 My Attendance

- **Path:** /gate-keeper/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)
- **Pass:** List/detail still correct after F5; no console/API 500.

13.8 Announcement

13.8.1 Announcement

- **Path:** /gate-keeper/broadcast-list
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

13.9 Communication

13.9.1 Communication

- **Path:** /gate-keeper/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

13.10 Calendar

13.10.1 Calendar

- **Path:** /gate-keeper/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

13.11 Notifications

13.11.1 Notifications

- **Path:** /gate-keeper/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

13.12 Escalation Management

13.12.1 Escalation Management

- **Path:** /gate-keeper/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

13.13 Visitors

13.13.1 Visitors

- **Path:** /gate-keeper/visitors
- **Do:** Open **Visitors**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Gate Keeper lists; PRM Front Office collections
- **Pass:** List/detail still correct after F5; no console/API 500.

13.14 Inward / Outward

13.14.1 Inward / Outward

- **Path:** /gate-keeper/registers
- **Do:** Open **Inward / Outward** (/gate-keeper/registers). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

13.15 Handover

13.15.1 Handover

- **Path:** /gate-keeper/handover
- **Do:** Open **Handover** (/gate-keeper/handover). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

13.16 Vendors

13.16.1 Vendors

- **Path:** /gate-keeper/vendors
- **Do:** Open **Vendors**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.

- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 14 — Gate Keeper Manager

Login: (HR create staff login) / —

Modules in sidebar: 13

14.1 Assign Duty List

14.1.1 Assign Duty List

- **Path:** /gatekeeper-manager/assign-duty-list
- **Do:** Open **Assign Duty List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

14.2 Incidents Management List

14.2.1 Incidents Management List

- **Path:** /gatekeeper-manager/incidents-list
- **Do:** Open **Incidents Management List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

14.3 Attendance

14.3.1 My Attendance

- **Path:** /gatekeeper-manager/attendance/my-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** GET /api/v1/hr/attendance/mine (hr.self) — Linked HR employee punches only
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/hr/attendance/mine (hr.self)
- **Pass:** List/detail still correct after F5; no console/API 500.

14.3.2 Gatekeepers Attendance

- **Path:** /gatekeeper-manager/attendance/gatekeepers-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

14.4 Task Management

14.4.1 Assign Tasks

- **Path:** /gatekeeper-manager/task-management/assign-tasks
- **Do:** Open **Assign Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

14.4.2 My Tasks

- **Path:** /gatekeeper-manager/task-management/my-tasks
- **Do:** Open **My Tasks**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

14.5 Leave Request

14.5.1 My Leave Requests

- **Path:** /gatekeeper-manager/leave-request/my-requests
- **Do:** Open **My Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

14.5.2 Received Leave Requests

- **Path:** /gatekeeper-manager/leave-request/received
- **Do:** Open **Received Leave Requests**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

14.6 Announcement

14.6.1 Announcement

- **Path:** /gatekeeper-manager/gatekeeper-broadcast-list
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

14.7 Communication

14.7.1 Communication

- **Path:** /gatekeeper-manager/communication/inbox
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox
- **Pass:** List/detail still correct after F5; no console/API 500.

14.8 Calendar

14.8.1 Calendar

- **Path:** /gatekeeper-manager/academic-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events

- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

14.9 Notifications

14.9.1 Notifications

- **Path:** /gatekeeper-manager/notifications
- **Do:** Open **Notifications**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** GET /api/v1/notifications — Own inbox; mark-read updates server
- **Reflects to:** Same role after refresh; related roles that share GET /api/v1/notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

14.10 Escalation Management

14.10.1 Escalation Management

- **Path:** /gatekeeper-manager/escalation-management
- **Do:** Open **Escalation Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 15 — Canteen Manager

Login: (HR create staff login) / —

Modules in sidebar: 7

15.1 Café

15.1.1 Café

- **Path:** /canteen-manager/dashboard
- **Do:** Open **Café** (/canteen-manager/dashboard). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/canteen-* — Canteen Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/canteen-*
- **Pass:** List/detail still correct after F5; no console/API 500.

15.2 Menu Management

15.2.1 Menu Management

- **Path:** /canteen-manager/menu-management
- **Do:** Open **Menu Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/canteen-* — Canteen Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/canteen-*
- **Pass:** List/detail still correct after F5; no console/API 500.

15.3 Inventory Management

15.3.1 Inventory Management

- **Path:** /canteen-manager/inventory
- **Do:** Open **Inventory Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/canteen-* — Canteen Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/canteen-*
- **Pass:** List/detail still correct after F5; no console/API 500.

15.4 Orders

15.4.1 Orders

- **Path:** /canteen-manager/orders
- **Do:** Open **Orders**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/canteen-* — Canteen Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/canteen-*
- **Pass:** List/detail still correct after F5; no console/API 500.

15.5 Requests & Approvals

15.5.1 Requests & Approvals

- **Path:** /canteen-manager/requests-approvals
- **Do:** Open **Requests & Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/canteen-* — Canteen Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/canteen-*

- **Pass:** List/detail still correct after F5; no console/API 500.

15.6 Reports

15.6.1 Reports

- **Path:** /canteen-manager/reports
- **Do:** Open **Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/canteen-* — Canteen Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/canteen-*
- **Pass:** List/detail still correct after F5; no console/API 500.

15.7 Announcement

15.7.1 Announcement

- **Path:** /canteen-manager/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 16 — IT Support Manager

Login: (HR create staff login) / —

Modules in sidebar: 9

16.1 Dashboard

16.1.1 Dashboard

- **Path:** /it-support-manager/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*
- **Pass:** List/detail still correct after F5; no console/API 500.

16.2 Asset Management

16.2.1 Asset Management

- **Path:** /it-support-manager/asset-management
- **Do:** Open **Asset Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*
- **Pass:** List/detail still correct after F5; no console/API 500.

16.3 Maintenance Inventory

16.3.1 Maintenance Inventory

- **Path:** /it-support-manager/maintenance-inventory
- **Do:** Open **Maintenance Inventory** (/it-support-manager/maintenance-inventory). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*
- **Pass:** List/detail still correct after F5; no console/API 500.

16.4 Tickets

16.4.1 Tickets

- **Path:** /it-support-manager/support-tickets
- **Do:** Open **Tickets**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*
- **Pass:** List/detail still correct after F5; no console/API 500.

16.5 Requests & Approvals

16.5.1 Requests & Approvals

- **Path:** /it-support-manager/requests-approvals
- **Do:** Open **Requests & Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*

- **Pass:** List/detail still correct after F5; no console/API 500.

16.6 Reports

16.6.1 Reports

- **Path:** /it-support-manager/reports
- **Do:** Open **Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*
- **Pass:** List/detail still correct after F5; no console/API 500.

16.7 Announcement

16.7.1 Announcement

- **Path:** /it-support-manager/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

16.8 Data Import

16.8.1 Data Import

- **Path:** /it-support-manager/data-import
- **Do:** Open **Data Import** (/it-support-manager/data-import). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*
- **Pass:** List/detail still correct after F5; no console/API 500.

16.9 Purchase Workflow

16.9.1 Purchase Workflow

- **Path:** /it-support-manager/purchase-workflow
- **Do:** Open **Purchase Workflow** (/it-support-manager/purchase-workflow). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/it-* — IT Support Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/it-*
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 17 — Stationery Store Manager

Login: (HR create staff login) / —

Modules in sidebar: 10

17.1 Dashboard

17.1.1 Dashboard

- **Path:** /stationery-store-manager/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

17.2 Stores

17.2.1 Stores

- **Path:** /stationery-store-manager/inventory
- **Do:** Open **Stores**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

17.3 Issue & Returns

17.3.1 Issue & Returns

- **Path:** /stationery-store-manager/issue-returns
- **Do:** Open **Issue & Returns**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

17.4 Requests & Approvals

17.4.1 Requests & Approvals

- **Path:** /stationery-store-manager/requests-approvals
- **Do:** Open **Requests & Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

17.5 Reports

17.5.1 Reports

- **Path:** /stationery-store-manager/reports
- **Do:** Open **Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*

- **Pass:** List/detail still correct after F5; no console/API 500.

17.6 Announcement

17.6.1 Announcement

- **Path:** /stationery-store-manager/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

17.7 Requirements

17.7.1 Requirements

- **Path:** /stationery-store-manager/requirements
- **Do:** Open **Requirements**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

17.8 Stock Issue

17.8.1 Stock Issue

- **Path:** /stationery-store-manager/stock-issue
- **Do:** Open **Stock Issue** (/stationery-store-manager/stock-issue). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

17.9 Critical Alerts

17.9.1 Critical Alerts

- **Path:** /stationery-store-manager/critical-alerts
- **Do:** Open **Critical Alerts**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

17.10 Purchase Workflow

17.10.1 Purchase Workflow

- **Path:** /stationery-store-manager/purchase-workflow
- **Do:** Open **Purchase Workflow** (/stationery-store-manager/purchase-workflow). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/stationery-* — Stationery Manager SoT
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/stationery-*
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 18 — Housekeeping Manager

Login: (HR create staff login) / —

Modules in sidebar: 13

18.1 Dashboard

18.1.1 Dashboard

- **Path:** /housekeeping-manager/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.2 Task Management

18.2.1 Task Management

- **Path:** /housekeeping-manager/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

18.3 Inventory

18.3.1 Inventory

- **Path:** /housekeeping-manager/inventory
- **Do:** Open **Inventory** (/housekeeping-manager/inventory). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.4 Requests & Approvals

18.4.1 Requests & Approvals

- **Path:** /housekeeping-manager/requests-approvals
- **Do:** Open **Requests & Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.5 Reports

18.5.1 Reports

- **Path:** /housekeeping-manager/reports
- **Do:** Open **Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory

- **Pass:** List/detail still correct after F5; no console/API 500.

18.6 Announcement

18.6.1 Announcement

- **Path:** /housekeeping-manager/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

18.7 Duty Allotment

18.7.1 Duty Allotment

- **Path:** /housekeeping-manager/duty-allotment
- **Do:** Open **Duty Allotment** (/housekeeping-manager/duty-allotment). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/gate-* (+ FO gate passes) — Gate/GKM write; PRM/Principal may see passes
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/gate-* (+ FO gate passes)
- **Pass:** List/detail still correct after F5; no console/API 500.

18.8 Schedules

18.8.1 Schedules

- **Path:** /housekeeping-manager/schedules
- **Do:** Open **Schedules**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.9 Star of the Month

18.9.1 Star of the Month

- **Path:** /housekeeping-manager/star-of-the-month
- **Do:** Open **Star of the Month** (/housekeeping-manager/star-of-the-month). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.10 RO Testing

18.10.1 RO Testing

- **Path:** /housekeeping-manager/ro-testing
- **Do:** Open **RO Testing** (/housekeeping-manager/ro-testing). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.11 Lost & Found

18.11.1 Lost & Found

- **Path:** /housekeeping-manager/lost-found
- **Do:** Open **Lost & Found** (/housekeeping-manager/lost-found). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.12 Inventory Requirement

18.12.1 Inventory Requirement

- **Path:** /housekeeping-manager/inventory-requirement
- **Do:** Open **Inventory Requirement** (/housekeeping-manager/inventory-requirement). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

18.13 Purchase Workflow

18.13.1 Purchase Workflow

- **Path:** /housekeeping-manager/purchase-workflow
- **Do:** Open **Purchase Workflow** (/housekeeping-manager/purchase-workflow). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/hk-* / shared-inventory — HK Manager; stores shared inventory
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/hk-* / shared-inventory
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 19 — Transport Manager

Login: (HR create staff login) / —

Modules in sidebar: 12

19.1 Dashboard

19.1.1 Dashboard

- **Path:** /transport-manager/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.2 Driver Management

19.2.1 Driver Management

- **Path:** /transport-manager/driver-management
- **Do:** Open **Driver Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.3 Vehicle Management

19.3.1 Vehicle Management

- **Path:** /transport-manager/vehicle-management
- **Do:** Open **Vehicle Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.4 Route Management

19.4.1 Route Management

- **Path:** /transport-manager/route-management
- **Do:** Open **Route Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.5 Route Data

19.5.1 Route Data

- **Path:** /transport-manager/route-data
- **Do:** Open **Route Data** (/transport-manager/route-data). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities

- **Pass:** List/detail still correct after F5; no console/API 500.

19.6 Student Transport

19.6.1 Student Transport

- **Path:** /transport-manager/student-transport
- **Do:** Open **Student Transport** (/transport-manager/student-transport). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.7 Assign Duty

19.7.1 Assign Duty

- **Path:** /transport-manager/assign-duty
- **Do:** Open **Assign Duty** (/transport-manager/assign-duty). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.8 Vehicle Maintenance

19.8.1 Vehicle Maintenance

- **Path:** /transport-manager/vehicle-maintenance
- **Do:** Open **Vehicle Maintenance** (/transport-manager/vehicle-maintenance). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.9 Request Approvals

19.9.1 Request Approvals

- **Path:** /transport-manager/request-approvals
- **Do:** Open **Request Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

19.10 Transport Expenses

19.10.1 Transport Expenses

- **Path:** /transport-manager/transport-expenses
- **Do:** Open **Transport Expenses**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities

- **Pass:** List/detail still correct after F5; no console/API 500.

19.11 Leave Request

19.11.1 Leave Request

- **Path:** /transport-manager/leave-request
- **Do:** File a leave/permission request; note ID; check Received on approver login.
- **Stored in:** /api/v1/leave-requests — Approver Received list; HR/Principal/Director as chain
- **Reflects to:** Approver Received Leave; notifications
- **Pass:** List/detail still correct after F5; no console/API 500.

19.12 Building Maintenance

19.12.1 Building Maintenance

- **Path:** /transport-manager/building-maintenance
- **Do:** Open **Building Maintenance** (/transport-manager/building-maintenance). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/activities — Admin/Principal write; others read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/activities
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 20 — Driver

Login: (HR create staff login) / —

Modules in sidebar: 5

20.1 Vehicle Management

20.1.1 Vehicle Details

- **Path:** /driver/vehicle-management/vehicle-details
- **Do:** Open **Vehicle Details**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/transport-* — TM/Driver; Track Bus also uses ProTrack
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/transport-*
- **Pass:** List/detail still correct after F5; no console/API 500.

20.1.2 Vehicle Documents

- **Path:** /driver/vehicle-management/vehicle-documents
- **Do:** Open **Vehicle Documents**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/documents/* — Student/employee folders; authorised download
- **Reflects to:** Same role after refresh; related roles that share /api/v1/documents/*
- **Pass:** List/detail still correct after F5; no console/API 500.

20.1.3 Vehicle Health Status

- **Path:** /driver/vehicle-management/vehicle-health-status
- **Do:** Open **Vehicle Health Status**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/transport-* — TM/Driver; Track Bus also uses ProTrack
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/transport-*
- **Pass:** List/detail still correct after F5; no console/API 500.

20.2 Attendance Management

20.2.1 Student Attendance

- **Path:** /driver/attendance-management/student-attendance
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/ops/transport-* — TM/Driver; Track Bus also uses ProTrack
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

20.2.2 Attendance History

- **Path:** /driver/attendance-management/attendance-history
- **Do:** Open attendance; mark/save if allowed; refresh; confirm persisted.
- **Stored in:** /api/v1/ops/transport-* — TM/Driver; Track Bus also uses ProTrack
- **Reflects to:** Parent/Student attendance views; Admin attendance read
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 21 — Joint Director

Login: (HR create staff login) / —

Modules in sidebar: 14

21.1 Dashboard

21.1.1 Dashboard

- **Path:** /joint-director/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/ops/jd-* (+ HR read) — Approvals/escalations/meetings/assets
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/jd-* (+ HR read)
- **Pass:** List/detail still correct after F5; no console/API 500.

21.2 Task Management

21.2.1 Task Management

- **Path:** /joint-director/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

21.3 Employee Management

21.3.1 JD Assistant

- **Path:** /joint-director/employee-management/jd-assistant
- **Do:** Open **JD Assistant** (/joint-director/employee-management/jd-assistant). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

21.3.2 Canteen Manager

- **Path:** /joint-director/employee-management/canteen-manager
- **Do:** Open **Canteen Manager** (/joint-director/employee-management/canteen-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

21.3.3 Store Manager

- **Path:** /joint-director/employee-management/store-manager
- **Do:** Open **Store Manager** (/joint-director/employee-management/store-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

21.3.4 IT Team Manager

- **Path:** /joint-director/employee-management/it-team-manager
- **Do:** Open **IT Team Manager** (/joint-director/employee-management/it-team-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

21.3.5 Transport Manager

- **Path:** /joint-director/employee-management/transport-manager
- **Do:** Open **Transport Manager** (/joint-director/employee-management/transport-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

21.3.6 House Keeping Manager

- **Path:** /joint-director/employee-management/housekeeping-manager
- **Do:** Open **House Keeping Manager** (/joint-director/employee-management/housekeeping-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

21.3.7 Drivers List

- **Path:** /joint-director/employee-management/drivers
- **Do:** Open **Drivers List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

21.4 Request Approvals

21.4.1 Request Approvals

- **Path:** /joint-director/request-approvals
- **Do:** Open **Request Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/jd-* (+ HR read) — Approvals/escalations/meetings/assets
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/jd-* (+ HR read)
- **Pass:** List/detail still correct after F5; no console/API 500.

21.5 Escalations

21.5.1 Escalations

- **Path:** /joint-director/escalations
- **Do:** Open **Escalations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations

- **Pass:** List/detail still correct after F5; no console/API 500.

21.6 Meetings & Calendar

21.6.1 Meetings & Calendar

- **Path:** /joint-director/meetings-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar
- **Pass:** List/detail still correct after F5; no console/API 500.

21.7 Assets & Inventory Overview

21.7.1 Assets & Inventory Overview

- **Path:** /joint-director/assets-inventory
- **Do:** Open **Assets & Inventory Overview** (/joint-director/assets-inventory). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/jd-* (+ HR read) — Approvals/escalations/meetings/assets
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/jd-* (+ HR read)
- **Pass:** List/detail still correct after F5; no console/API 500.

21.8 Announcement

21.8.1 Announcement

- **Path:** /joint-director/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 22 — Joint Director Assistant

Login: (HR create staff login) / —

Modules in sidebar: 13

22.1 Dashboard

22.1.1 Dashboard

- **Path:** /joint-director-assistant/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/ops/jd-* (+ HR read) — Approvals/escalations/meetings/assets
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/jd-* (+ HR read)
- **Pass:** List/detail still correct after F5; no console/API 500.

22.2 Task Management

22.2.1 Task Management

- **Path:** /joint-director-assistant/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** Same role after refresh; related roles that share /api/v1/tasks
- **Pass:** List/detail still correct after F5; no console/API 500.

22.3 Employee Management

22.3.1 Canteen Manager

- **Path:** /joint-director-assistant/employee-management/canteen-manager
- **Do:** Open **Canteen Manager** (/joint-director-assistant/employee-management/canteen-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

22.3.2 Store Manager

- **Path:** /joint-director-assistant/employee-management/store-manager
- **Do:** Open **Store Manager** (/joint-director-assistant/employee-management/store-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

22.3.3 IT Team Manager

- **Path:** /joint-director-assistant/employee-management/it-team-manager
- **Do:** Open **IT Team Manager** (/joint-director-assistant/employee-management/it-team-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

22.3.4 Transport Manager

- **Path:** /joint-director-assistant/employee-management/transport-manager
- **Do:** Open **Transport Manager** (/joint-director-assistant/employee-management/transport-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

22.3.5 House Keeping Manager

- **Path:** /joint-director-assistant/employee-management/housekeeping-manager
- **Do:** Open **House Keeping Manager** (/joint-director-assistant/employee-management/housekeeping-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

22.3.6 Drivers List

- **Path:** /joint-director-assistant/employee-management/drivers
- **Do:** Open **Drivers List**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** Same role after refresh; related roles that share /api/v1/hr/*
- **Pass:** List/detail still correct after F5; no console/API 500.

22.4 Request Approvals

22.4.1 Request Approvals

- **Path:** /joint-director-assistant/request-approvals
- **Do:** Open **Request Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/ops/jd-* (+ HR read) — Approvals/escalations/meetings/assets
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/jd-* (+ HR read)
- **Pass:** List/detail still correct after F5; no console/API 500.

22.5 Escalations

22.5.1 Escalations

- **Path:** /joint-director-assistant/escalations
- **Do:** Open **Escalations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** Same role after refresh; related roles that share /api/v1/escalations
- **Pass:** List/detail still correct after F5; no console/API 500.

22.6 Meetings & Calendar

22.6.1 Meetings & Calendar

- **Path:** /joint-director-assistant/meetings-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** Same role after refresh; related roles that share GET/PUT /api/v1/calendar

- **Pass:** List/detail still correct after F5; no console/API 500.

22.7 Assets & Inventory Overview

22.7.1 Assets & Inventory Overview

- **Path:** /joint-director-assistant/assets-inventory
- **Do:** Open **Assets & Inventory Overview** (/joint-director-assistant/assets-inventory). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/ops/jd-* (+ HR read) — Approvals/escalations/meetings/assets
- **Reflects to:** Same role after refresh; related roles that share /api/v1/ops/jd-* (+ HR read)
- **Pass:** List/detail still correct after F5; no console/API 500.

22.8 Announcement

22.8.1 Announcement

- **Path:** /joint-director-assistant/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** Same role after refresh; related roles that share GET/POST /api/v1/announcements
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 23 — Joint Director Audit

Login: (HR create staff login) / —

Modules in sidebar: 29

23.1 Dashboard

23.1.1 Dashboard

- **Path:** /joint-director-audit/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2 Audit Configuration

23.2.1 Audit Templates

- **Path:** /joint-director-audit/audit-configuration/audit-templates
- **Do:** Open **Audit Templates**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role; Process/Quality Execute Audit checklist
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.2 Checklist Sections

- **Path:** /joint-director-audit/audit-configuration/checklist-sections
- **Do:** Open **Checklist Sections**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.3 Question Bank

- **Path:** /joint-director-audit/audit-configuration/question-bank
- **Do:** Open **Question Bank** (/joint-director-audit/audit-configuration/question-bank). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.4 Response Types

- **Path:** /joint-director-audit/audit-configuration/response-types
- **Do:** Open **Response Types**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.5 Scoring Rules

- **Path:** /joint-director-audit/audit-configuration/scoring-rules

- **Do:** Open **Scoring Rules**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.6 Workflow Rules

- **Path:** /joint-director-audit/audit-configuration/workflow-rules
- **Do:** Open **Workflow Rules**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.7 Visibility Rules

- **Path:** /joint-director-audit/audit-configuration/visibility-rules
- **Do:** Open **Visibility Rules**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.8 Template Versioning

- **Path:** /joint-director-audit/audit-configuration/template-versioning
- **Do:** Open **Template Versioning** (/joint-director-audit/audit-configuration/template-versioning). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role; Process/Quality Execute Audit checklist
- **Pass:** List/detail still correct after F5; no console/API 500.

23.2.9 Publish Templates

- **Path:** /joint-director-audit/audit-configuration/publish-templates
- **Do:** Open **Publish Templates**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role; Process/Quality Execute Audit checklist
- **Pass:** List/detail still correct after F5; no console/API 500.

23.3 Audit Planning

23.3.1 Audit Planning

- **Path:** /joint-director-audit/audit-planning
- **Do:** Open **Audit Planning** (/joint-director-audit/audit-planning). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.4 Audit Assignment

23.4.1 Assign Audits

- **Path:** /joint-director-audit/audit-assignment/assign-audits
- **Do:** Open **Assign Audits**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.4.2 Assignment History

- **Path:** /joint-director-audit/audit-assignment/assignment-history
- **Do:** Open **Assignment History** (/joint-director-audit/audit-assignment/assignment-history). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.4.3 Workload Overview

- **Path:** /joint-director-audit/audit-assignment/workload-overview
- **Do:** Open **Workload Overview** (/joint-director-audit/audit-assignment/workload-overview). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.4.4 Reassign Audit

- **Path:** /joint-director-audit/audit-assignment/reassign-audit
- **Do:** Open **Reassign Audit** (/joint-director-audit/audit-assignment/reassign-audit). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.5 Audit Monitoring

23.5.1 Audit Monitoring

- **Path:** /joint-director-audit/audit-monitoring
- **Do:** Open **Audit Monitoring** (/joint-director-audit/audit-monitoring). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.6 Findings & Compliance

23.6.1 Findings & Compliance

- **Path:** /joint-director-audit/findings-compliance
- **Do:** Open **Findings & Compliance** (/joint-director-audit/findings-compliance). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates

- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.7 Reports & Analytics

23.7.1 Reports & Analytics

- **Path:** /joint-director-audit/reports-analytics
- **Do:** Open **Reports & Analytics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.8 Task Management

23.8.1 Task Management

- **Path:** /joint-director-audit/task-management/my-tasks
- **Do:** Open **Task Management**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/tasks — Assignee My Tasks; assigner can track
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.9 Employee Management

23.9.1 HR Manager

- **Path:** /joint-director-audit/employee-management/hr-manager
- **Do:** Open **HR Manager** (/joint-director-audit/employee-management/hr-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.9.2 HR Executive

- **Path:** /joint-director-audit/employee-management/hr-executive
- **Do:** Open **HR Executive** (/joint-director-audit/employee-management/hr-executive). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.9.3 Process Audit Manager

- **Path:** /joint-director-audit/employee-management/process-audit-manager
- **Do:** Open **Process Audit Manager** (/joint-director-audit/employee-management/process-audit-manager). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.9.4 Process Audit Executive

- **Path:** /joint-director-audit/employee-management/process-audit-executive
- **Do:** Open **Process Audit Executive** (/joint-director-audit/employee-management/process-audit-executive). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.9.5 Quality Audit Principal

- **Path:** /joint-director-audit/employee-management/quality-audit-principal
- **Do:** Open **Quality Audit Principal** (/joint-director-audit/employee-management/quality-audit-principal). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.9.6 Quality Audit Executive

- **Path:** /joint-director-audit/employee-management/quality-audit-executive
- **Do:** Open **Quality Audit Executive** (/joint-director-audit/employee-management/quality-audit-executive). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/hr/* — Staff directory; links logins; JD employee overview read
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.10 Request Approvals

23.10.1 Request Approvals

- **Path:** /joint-director-audit/request-approvals
- **Do:** Open **Request Approvals**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/audit-* — JD Audit config; Process/Quality execute against templates
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.11 Escalations

23.11.1 Escalations

- **Path:** /joint-director-audit/escalations
- **Do:** Open **Escalations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

23.12 Meetings & Calendar

23.12.1 Meetings & Calendar

- **Path:** /joint-director-audit/meetings-calendar
- **Do:** Open calendar; confirm seeded/campus events appear. Editors: add one event and refresh.
- **Stored in:** GET/PUT /api/v1/calendar — All roles with calendar.read see events
- **Reflects to:** JD Audit monitoring/findings; paired auditor role

- **Pass:** List/detail still correct after F5; no console/API 500.

23.13 Announcement

23.13.1 Announcement

- **Path:** /joint-director-audit/broadcast
- **Do:** Open list; if you can create, send to a known audience role; switch login and confirm it appears.
- **Stored in:** GET/POST /api/v1/announcements — Audience roles see list; acknowledgements per login
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 24 — Process Auditor

Login: (HR create staff login) / —

Modules in sidebar: 17

24.1 Dashboard

24.1.1 Dashboard

- **Path:** /process-auditor/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.2 Audit Management

24.2.1 My Audits

- **Path:** /process-auditor/audit-management/my-audits
- **Do:** Open **My Audits**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.2.2 Audit Schedule

- **Path:** /process-auditor/audit-management/audit-schedule
- **Do:** Open **Audit Schedule** (/process-auditor/audit-management/audit-schedule). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.2.3 Audit History

- **Path:** /process-auditor/audit-management/audit-history
- **Do:** Open **Audit History** (/process-auditor/audit-management/audit-history). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.3 Observations

24.3.1 Open Observations

- **Path:** /process-auditor/observations/open
- **Do:** Open **Open Observations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.3.2 Closed Observations

- **Path:** /process-auditor/observations/closed
- **Do:** Open **Closed Observations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.4 Corrective Actions

24.4.1 Escalations

- **Path:** /process-auditor/corrective-actions/escalations
- **Do:** Open **Escalations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.4.2 Root Cause Analysis

- **Path:** /process-auditor/corrective-actions/root-cause-analysis
- **Do:** Open **Root Cause Analysis**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.4.3 Action Taken Reports

- **Path:** /process-auditor/corrective-actions/action-taken-reports
- **Do:** Open **Action Taken Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.5 Reports

24.5.1 Audit Reports

- **Path:** /process-auditor/reports/audit-reports
- **Do:** Open **Audit Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.5.2 Observation Reports

- **Path:** /process-auditor/reports/observation-reports
- **Do:** Open **Observation Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.5.3 Compliance Reports

- **Path:** /process-auditor/reports/compliance-reports
- **Do:** Open **Compliance Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.5.4 Pending Actions

- **Path:** /process-auditor/reports/pending-actions
- **Do:** Open **Pending Actions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.5.5 Closed Actions

- **Path:** /process-auditor/reports/closed-actions
- **Do:** Open **Closed Actions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.6 Communication

24.6.1 Communication

- **Path:** /process-auditor/communication
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox; JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.7 Components & Rubrics

24.7.1 Components & Rubrics

- **Path:** /process-auditor/components
- **Do:** Open **Components & Rubrics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

24.8 Shared Deviations

24.8.1 Shared Deviations

- **Path:** /process-auditor/shared-deviations
- **Do:** Open **Shared Deviations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.

- **Stored in:** /api/v1/campus-audit/process-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

Step 25 — Quality Auditor

Login: (HR create staff login) / —

Modules in sidebar: 17

25.1 Dashboard

25.1.1 Dashboard

- **Path:** /quality-auditor/dashboard
- **Do:** Open dashboard; confirm KPIs/cards load (not blank). Note counts for later checks.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.2 Audit Management

25.2.1 My Audits

- **Path:** /quality-auditor/audit-management/my-audits
- **Do:** Open **My Audits**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.2.2 Audit Schedule

- **Path:** /quality-auditor/audit-management/audit-schedule
- **Do:** Open **Audit Schedule** (/quality-auditor/audit-management/audit-schedule). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.2.3 Audit History

- **Path:** /quality-auditor/audit-management/audit-history
- **Do:** Open **Audit History** (/quality-auditor/audit-management/audit-history). Exercise primary action (view/list/add). Refresh and confirm data behaviour.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.3 Observations

25.3.1 Open Observations

- **Path:** /quality-auditor/observations/open
- **Do:** Open **Open Observations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.3.2 Closed Observations

- **Path:** /quality-auditor/observations/closed
- **Do:** Open **Closed Observations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.4 Corrective Actions

25.4.1 Escalations

- **Path:** /quality-auditor/corrective-actions/escalations
- **Do:** Open **Escalations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/escalations — Next level in hierarchy; notifications
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.4.2 Root Cause Analysis

- **Path:** /quality-auditor/corrective-actions/root-cause-analysis
- **Do:** Open **Root Cause Analysis**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.4.3 Action Taken Reports

- **Path:** /quality-auditor/corrective-actions/action-taken-reports
- **Do:** Open **Action Taken Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.5 Reports

25.5.1 Audit Reports

- **Path:** /quality-auditor/reports/audit-reports
- **Do:** Open **Audit Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.5.2 Observation Reports

- **Path:** /quality-auditor/reports/observation-reports
- **Do:** Open **Observation Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.5.3 Compliance Reports

- **Path:** /quality-auditor/reports/compliance-reports
- **Do:** Open **Compliance Reports**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.5.4 Pending Actions

- **Path:** /quality-auditor/reports/pending-actions
- **Do:** Open **Pending Actions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.5.5 Closed Actions

- **Path:** /quality-auditor/reports/closed-actions
- **Do:** Open **Closed Actions**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.6 Communication

25.6.1 Communication

- **Path:** /quality-auditor/communication
- **Do:** Open inbox; send a short message to another campus login; confirm thread on recipient.
- **Stored in:** /api/v1/communication — Recipient inbox; campus logins as directory
- **Reflects to:** Recipient Communication inbox; JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.7 Components & Rubrics

25.7.1 Components & Rubrics

- **Path:** /quality-auditor/components
- **Do:** Open **Components & Rubrics**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.
- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

25.8 Shared Deviations

25.8.1 Shared Deviations

- **Path:** /quality-auditor/shared-deviations
- **Do:** Open **Shared Deviations**. If Add exists: create one record with today's date in a field. Hard refresh (F5). Confirm the row is still there.

- **Stored in:** /api/v1/campus-audit/quality-* — Auditor execution; JD Audit monitors findings
- **Reflects to:** JD Audit monitoring/findings; paired auditor role
- **Pass:** List/detail still correct after F5; no console/API 500.

ProTrack365 bus tracking (deep check)

- backend/.env: PROTRACK_ADMIN_* and PROTRACK_PARENT_* set; **restart API**.
- GET /api/v1/protrack/status → configured: true.
- GET /api/v1/protrack/devices?panel=admin → device list (if ProTrack returns permission denied, enable Open API on the ProTrack account).
- Transport Manager → Route Data → Track → select IMEI → live lat/lng on map.
- Parent → Transport → Track Bus → parent panel devices.

Biometric (deep check)

· Item · Expect ·

· Teacher Biometric → Sync · Rows update; F5 keeps them (/ops/biometric-attendance) ·

· Teacher override · Final status + reason kept after F5 ·

· Coordinator Biometric · Can view/sync; override may be disabled ·

· Live eSSL/RFID device · **Not integrated yet** (RFID_IMPORT_ENABLED=false) — Sync is simulated ·

End-to-end campus story (optional one-pass)

- **Admin** — class + section; confirm teacher login exists.
- **HR** — ensure teacher employee linked to login.
- **PRM** — enquiry → enroll student + parent password.
- **Account Head** — bootstrap ledger if needed; collect one fee.
- **Principal** — teacher allocation + class timetable; F5 persists.
- **Teacher** — mark attendance; submit lesson plan; biometric sync.
- **Coordinator** — approve lesson plan; view biometric.
- **Parent** — select child; see timetable/attendance/results; Track Bus.
- **Librarian** — book list hydrate; optional issue.
- **Gate** — add visitor; F5 persists.
- **Transport** — vehicles + Track Bus (ProTrack).
- **JD Audit** — template; **Process Auditor** — observation; F5 persists.

Known gaps

- Live eSSL/RFID import not enabled.
- ProTrack Open API must be allowed on client accounts (else permission denied).
- Some ops screens are thin CRUD over JSON collections — still must survive refresh.
- SMTP / WhatsApp / live Razorpay may be demo stubs.